P5	**SLO4** **AC4.1** **Loops** **Mathematical Operators**	**P5 Binary Conversion Program** A program converts short binary patterns into a non-negative denary integer. A valid input must contain at most four binary digits. Each digit must be either 0 or 1. If the user enters an empty pattern, the program must end immediately. The program does not convert patterns longer than 4 bits and should provide the user with a suitable error message asking them to re-enter with a valid pattern. Create a function that checks whether every digit is either 0 or 1. The function should be used in your main program before any conversions. If all digits are valid, a multiplier is used to represent each binary place value, doubling after each digit is processed. • When the program encounters a 1, the corresponding place value is added to a running total. • When it encounters a 0, nothing is added.

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- Once all the digits have been processed, the denary result is displayed.

Your program must:

- Use clear and meaningful variable names.
- Include comments explaining the main parts of the program.
- Use a function to check that all digits are either 0 or 1
- Produce clear and helpful error messaging.

Test Data

You should use the data below to test your solution.
You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected.

Input	Expected output
1100	Displays 12
0000	Displays 0
11111	Displays and explains error
1012	Displays and explains error
Empty input	Program exits

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